

TPC-406: The Complete-Shell C1 Local-Entry Boundary

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Abstract

TPC-405 established an inverse-height bound for one locally normalized adjacent entry of a selected-prime CRT proxy, but its certificate used only small selected prefixes of the prime shell. We close that finite selection gap: for each of five kernel heights, we select every one of the 872 primes in $8192 < p \leq 16384$, impose the explicit alternating CRT residues, and reconstruct an origin above 10^6 . The same exact argument gives $0 \leq z \leq t_1/(a_{\min}\sqrt{S_0S_1}) \leq 4/(a_{\min}H) \leq 4/H$. An exact rational certificate and an independent literal masked-energy replay verify all five complete-shell rows. The normalized observations range from 0.0217719 at $H = 16$ to 0.0013396 at $H = 256$; these decimals are finite observations, not an asymptotic theorem. The result remains a bound for one synthetic proxy entry, not a complete operator, arithmetic, or twin-prime theorem.

1 Scope and complete-shell model

Let H, N, Q be integers with $H \geq 1$, $N \geq H + 2$, and $Q > N$. Assume the complete shell $(Q, 2Q]$ has even cardinality $2m$, and let $p_0 < \dots < p_{2m-1}$ be every prime in that shell. Impose

$$o \equiv 0 \pmod{p_i} \ (i \text{ even}), \quad o \equiv -N \pmod{p_i} \ (i \text{ odd}).$$

For the certificate, $Q = 8192$, the shell has 872 primes, and hence $m = 436$. A CRT representative o is chosen above 10^6 ; since $p_i > Q > N$, the residues determine the hits in the half-open window $I_o = \{o, \dots, o + N - 1\}$.

$$t_d = \frac{H^2}{H^2 + d^2}, \quad S_0 = \sum_{d=1}^{N-1} t_d^2, \quad S_1 = \sum_{d=1}^{N-2} t_d^2 + t_1^2,$$

and $a_i = p_i^3/[Q^2(p_i - 1)]$. Write

$$P_- = \sum_{i \text{ odd}} a_i, \quad V_- = \sum_{i \text{ odd}} a_i^2, \quad V_+ = \sum_{i \text{ even}} a_i^2.$$

The selected-prime local proxy inherited from TPC-404 has

$$G_0 = V_- S_0, \quad G_1 = V_- S_1 + V_+(S_1 - t_1^2), \quad M = t_1 P_-.$$

The word “complete” refers to the finite declared shell, not to the full operator or to an arithmetic source.

2 Complete-shell adjacent-entry theorem

Theorem. Put $a_{\min} = \min_i a_i$ and $z = M/\sqrt{G_0 G_1}$. Under the hypotheses above,

$$0 \leq z \leq \frac{t_1}{a_{\min} \sqrt{S_0 S_1}} \leq \frac{4}{a_{\min} H} \leq \frac{4}{H}. \quad (1)$$

Proof. All amplitudes and kernel terms are positive. Also $S_1 - t_1^2 = \sum_{d=1}^{N-2} t_d^2 \geq 0$, so $G_1 \geq V_- S_1$. Cauchy–Schwarz gives $P_-^2 \leq m V_-$, while the m odd-indexed terms give $V_- \geq m a_{\min}^2$. Consequently,

$$z^2 \leq \frac{t_1^2 m V_-}{(V_- S_0)(V_- S_1)} = \frac{t_1^2 m}{V_- S_0 S_1} \leq \frac{t_1^2}{a_{\min}^2 S_0 S_1}. \quad (2)$$

For $1 \leq d \leq H$, $t_d \geq 1/2$. The condition $N \geq H + 2$ places these terms in both finite sums, so $S_0, S_1 \geq H/4$. Finally $a_i = (p_i/Q)^2 p_i / (p_i - 1) > 1$ because $p_i > Q$. Taking square roots in (2) proves (1). $\square$

3 Exact complete-shell certificate

The certificate fixes $Q = 8192$, uses the complete shell of 872 primes, and sets $N = 4H$ for $H \in \{16, 32, 66, 128, 256\}$. Each row stores exact rational kernel sums, amplitudes, CRT residue and period, local energies, and both sides of (1). The independent checker rebuilds the sieve and CRT origin and then replays both masked row energies literally. Thus the complete-shell claim is not inferred from a small-prime prefix.

H	N	normalized entry (float64 observation)
16	64	0.021771929097350
32	128	0.010802825201831
66	264	0.005213845896508
128	512	0.002682397867489
256	1024	0.001339555645667

The product of the 872 shell primes has 3558 decimal digits, but all certificate quantities are handled as exact integers or rational numbers. The table is therefore a numerical observation extracted from exact squares; it is not used as a fit or as evidence for a growing operator theorem.

4 Route boundary

TPC-406 advances the finite local proxy question from a selected prefix to the complete declared shell. It still concerns one adjacent entry and one explicit alternating CRT profile. It does not bound all entries, the complete normalized matrix or operator, arbitrary origins, unselected source primes, the physical h_0 , or the arithmetic sign law. Accordingly it pays no arithmetic L^2 estimate, no fixed-power $1/400$ credit, and no Route-A, Route-B, or twin-prime conclusion.

complete-shell local-entry boundary	PROVED_EXACT_FINITE
five-row rational certificate	PROVED_EXACT_FINITE
decimal scale observations	NUMERICAL_OBSERVATION
full normalized operator theorem	OPEN
arithmetic advance / fixed-power credit	NO / 0
Route-A / Route-B / twin-prime result	OPEN / OPEN / NONE

The next scoped question is `TEST_C1_COMPLETE_SHELL_LOCAL_ENTRY_SCALE_EXTENSION`.

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